

Experiment No. 2

Aim:- To understand the SaaS delivery model and deploy a G Suite cloud-based application that can be accessed by end users through a web browser without local installation.

Theory –

SaaS is also known as "On-Demand Software". It is a software distribution model in which services are hosted by a cloud service provider. These services are available to end-users over the internet so, the end-users do not need to install any software on their devices to access these services.

There are the following services provided by SaaS providers -

Business Services - SaaS Provider provides various business services to start-up the business. The SaaS business services include ERP (Enterprise Resource Planning), CRM (Customer Relationship Management), billing, and sales.

Document Management - SaaS document management is a software application offered by a third party (SaaS providers) to create, manage, and track electronic documents.

Example: Slack, Samepage, Box, and Zoho Forms.

Social Networks - As we all know, social networking sites are used by the general public, so social networking service providers use SaaS for their convenience and handle the general public's information.

Mail Services - To handle the unpredictable number of users and load on e-mail services, many e-mail providers offering their services using SaaS.



Advantages of SaaS cloud computing layer

1) SaaS is easy to buy:

SaaS pricing is based on a monthly fee or annual fee subscription, so it allows organizations to access business functionality at a low cost, which is less than licensed applications.

Unlike traditional software, which is sold as a licensed based with an up-front cost (and often an optional ongoing support fee), SaaS providers are generally pricing the applications using a subscription fee, most commonly a monthly or annually fee.

2. One to Many:

SaaS services are offered as a one-to-many model means a single instance of the application is shared by multiple users.

3. Less hardware required for SaaS:

The software is hosted remotely, so organizations do not need to invest in additional hardware.

4. Low maintenance required for SaaS:

Software as a service removes the need for installation, set-up, and daily maintenance for the organizations. The initial set-up cost for SaaS is typically less than the enterprise software. SaaS vendors are pricing their applications based on some usage parameters, such as a number of users using the application. So, SaaS does easy to monitor and automatic updates.

5. No special software or hardware versions required:

All users will have the same version of the software and typically access it through the web browser. SaaS reduces IT support costs by outsourcing hardware and software maintenance and support to the IaaS provider.

6. Multidevice support:

SaaS services can be accessed from any device such as desktops, laptops, tablets, phones, and thin clients.

7. API Integration:

SaaS services easily integrate with other software or services through standard APIs.

8. No client-side installation:

SaaS services are accessed directly from the service provider using the internet connection, so do not need to require any software installation.

Disadvantages of SaaS cloud computing layer

1) Security:

Actually, data is stored in the cloud, so security may be an issue for some users. However, cloud computing is not more secure than in-house deployment.

2) Latency issue:

Since data and applications are stored in the cloud at a variable distance from the end-user, there is a possibility that there may be greater latency when interacting with the application compared to local deployment. Therefore, the SaaS model is not suitable for applications whose demand response time is in milliseconds.

3) Total Dependency on Internet:

Without an internet connection, most SaaS applications are not usable.

4) Switching between SaaS vendors is difficult:

Switching SaaS vendors involves the difficult and slow task of transferring the very large data files over the internet and then converting and importing them into another SaaS also.

Popular SaaS Providers:



The below table shows some popular SaaS providers and services that are provided by them

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Provider	Services
Salseforce.com	On-demand CRM solutions
Microsoft Office 365	Online office suite
Google Apps	Gmail, Google Calendar, Docs, and sites
NetSuite	ERP, accounting, order management, CRM, Professionals Services Automation (PSA), and e-commerce applications.
GoToMeeting	Online meeting and video-conferencing software
Constant Contact	E-mail marketing, online survey, and event marketing
Oracle CRM	CRM applications
Workday, Inc	Human capital management, payroll, and financial management.

Example:

Google Workspace (formerly known as Google Apps and later G Suite) is a collection of cloud computing, productivity and collaboration tools, software and products developed and marketed by Google. It was first launched in 2006 as Google Apps for Your Domain and rebranded as G Suite in 2016. Google Workspace consists of Gmail, Contacts, Calendar, Meet and Chat for communication; Currents for employee engagement; Drive for storage; and the Google Docs suite for content creation. An Admin Panel is provided for managing users and services. Depending on edition Google Workspace may also include the digital interactive whiteboard Jam board and an option to purchase such add-ons as the telephony service Voice. The education edition adds a learning platform Google Classroom and today has the name Workspace for Education.

While most of these services are individually available at no cost to consumers who use their free Google (Gmail) accounts, Google Workspace adds enterprise features such as custom email addresses at a domain (e.g., @yourcompany.com), an option for unlimited Drive storage, additional administrative tools and advanced settings, as well as 24/7 phone and email support

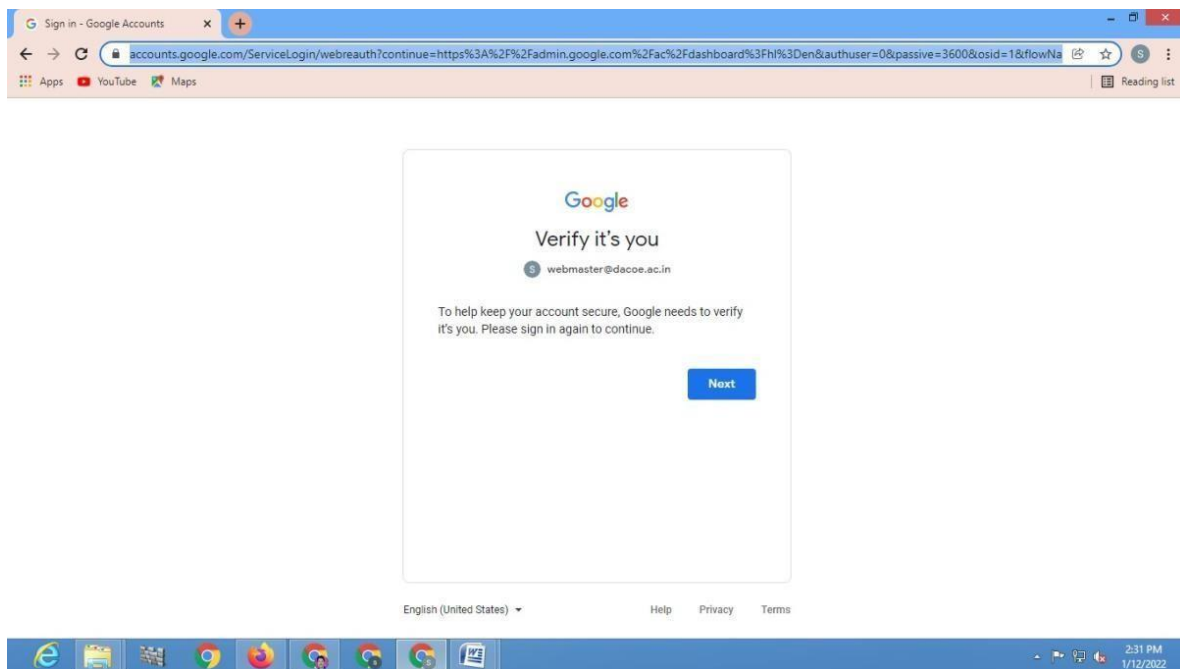
How to start using G Suite:

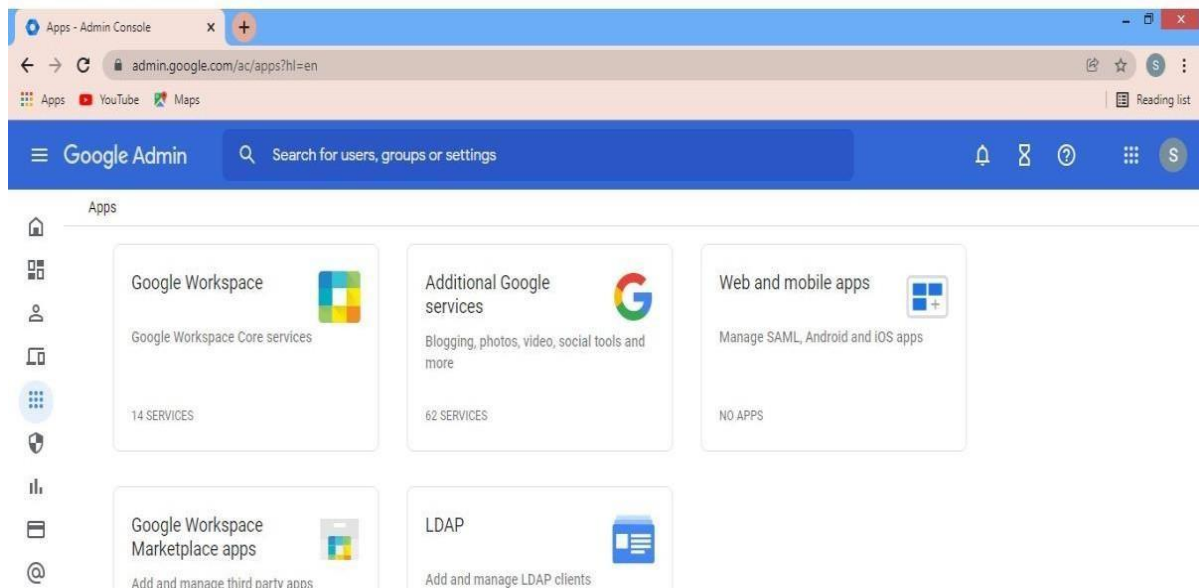
1) Sign in to your G Suite administrator account and set up your first users. You can manage all of the G Suite services for your business, including mobile device management, data migration, setting password requirements, and much more. This guide walks you through the admin sign-in process.

After you sign in, you will need to set up your G Suite account by

- 1.) [verifying](#) you own your domain,
- 2.) adding user accounts for your employees, and
- 3.) when all employees have been added, setting up Gmail as your email client by [updating your MX records](#). Don't worry, none of these changes will affect your website, if you have one.

Screenshots :





Conclusion: Thus, we have studied and implemented Software as a Service (**SaaS**).